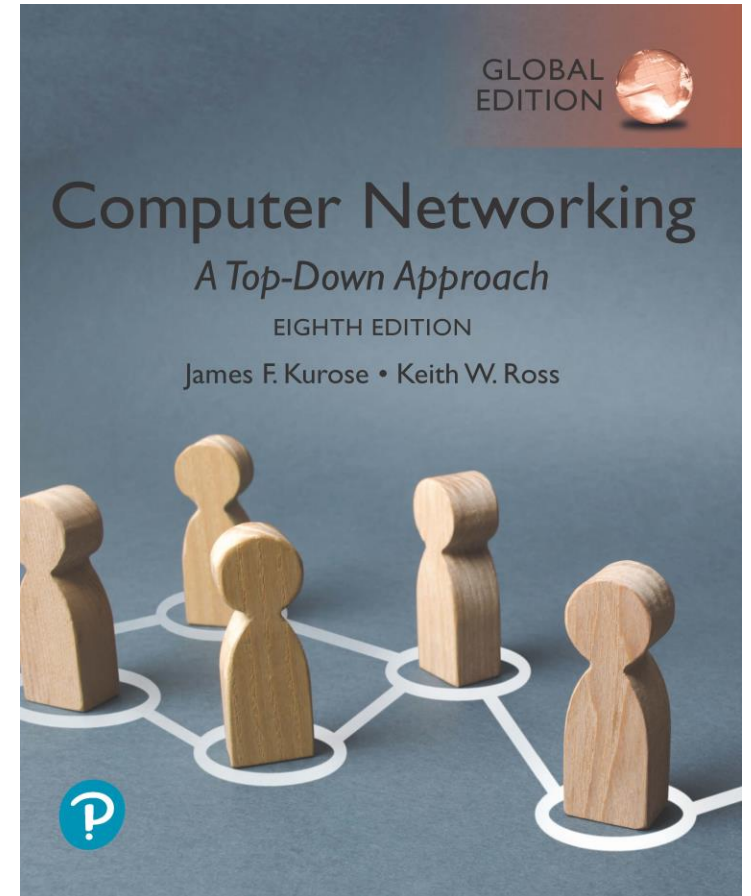


# Chapter 4

## Network Layer: Data Plane



### *Computer Networking: A Top-Down Approach*

8<sup>th</sup> Edition, Global Edition

Jim Kurose, Keith Ross

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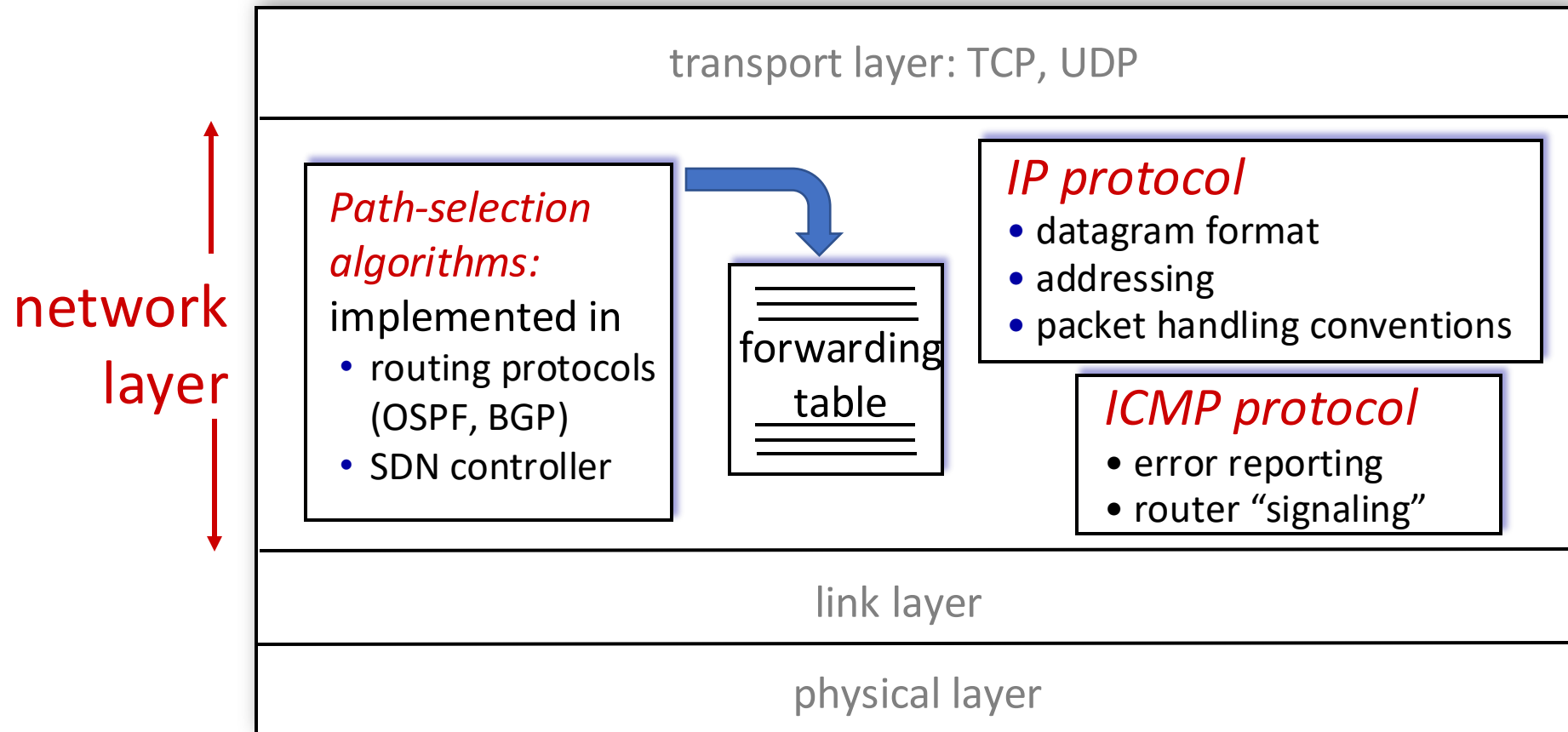
# Network layer: “data plane” roadmap

- Network layer: overview
  - data plane
  - control plane
- What’s inside a router
  - input ports, switching, output ports
  - buffer management, scheduling
- IP: the Internet Protocol
  - datagram format
  - addressing
  - network address translation
  - IPv6

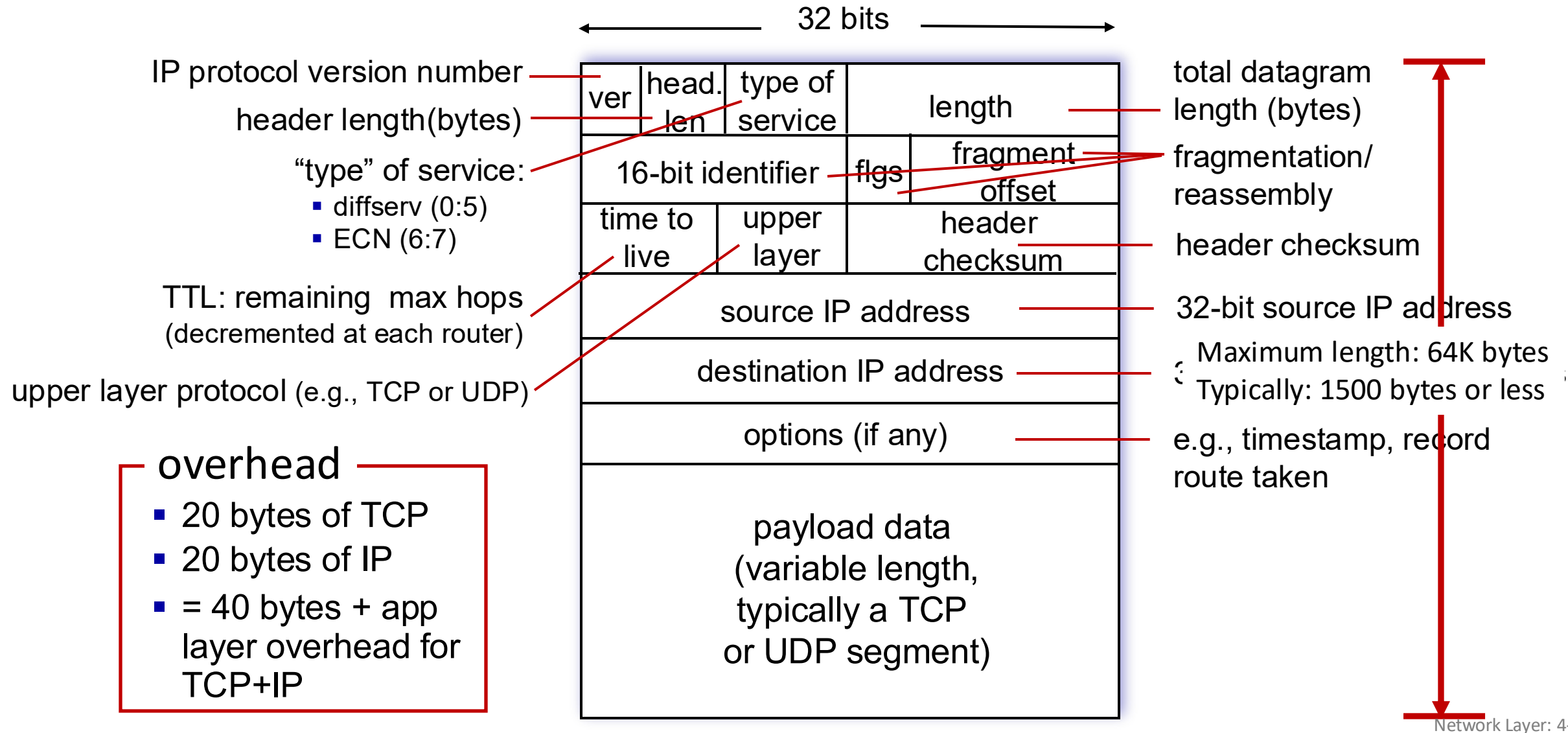


# Network Layer: Internet

host, router network layer functions:

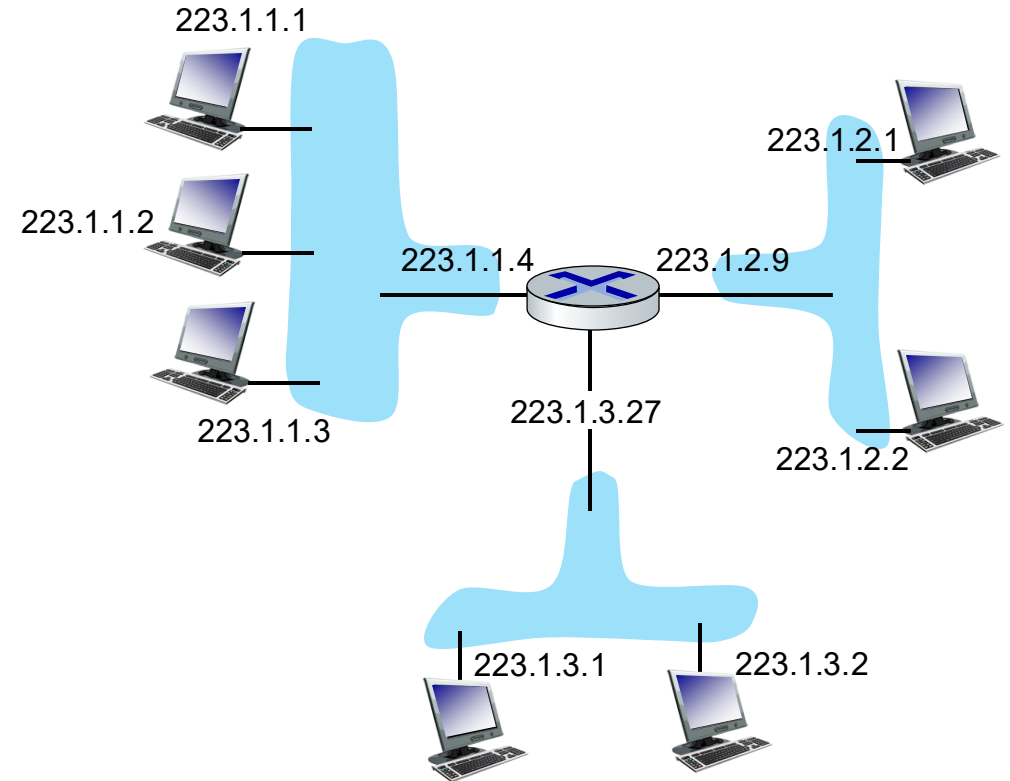


# IP Datagram format



# IP addressing: introduction

- **IP address:** 32-bit identifier associated with each host or router *interface*
- **interface:** connection between host/router and physical link
  - router's typically have multiple interfaces
  - host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)



dotted-decimal IP address notation:

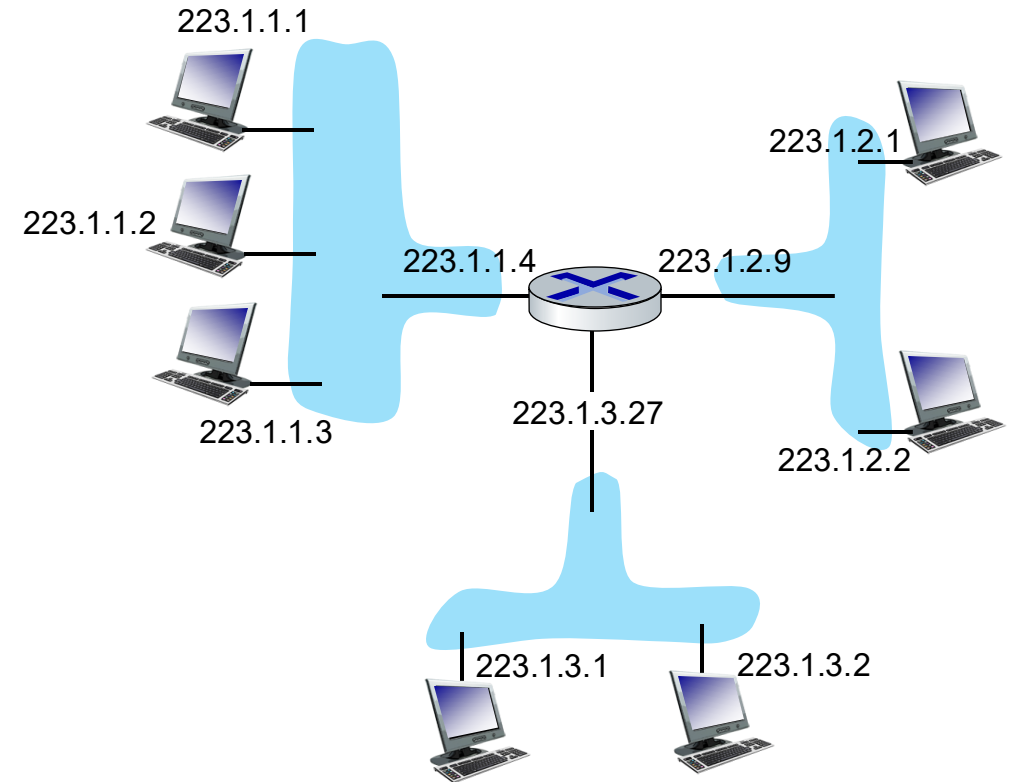
223.1.1.1 = 11011111 00000001 00000001 00000001

223                      1                      1                      1

Network Layer: 4-5

# IP addressing: introduction

- **IP address:** 32-bit identifier associated with each host or router *interface*
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dotted-decimal IP address notation:

223.1.1.1 = 11011111 00000001 00000001 00000001

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Network Layer: 4-6

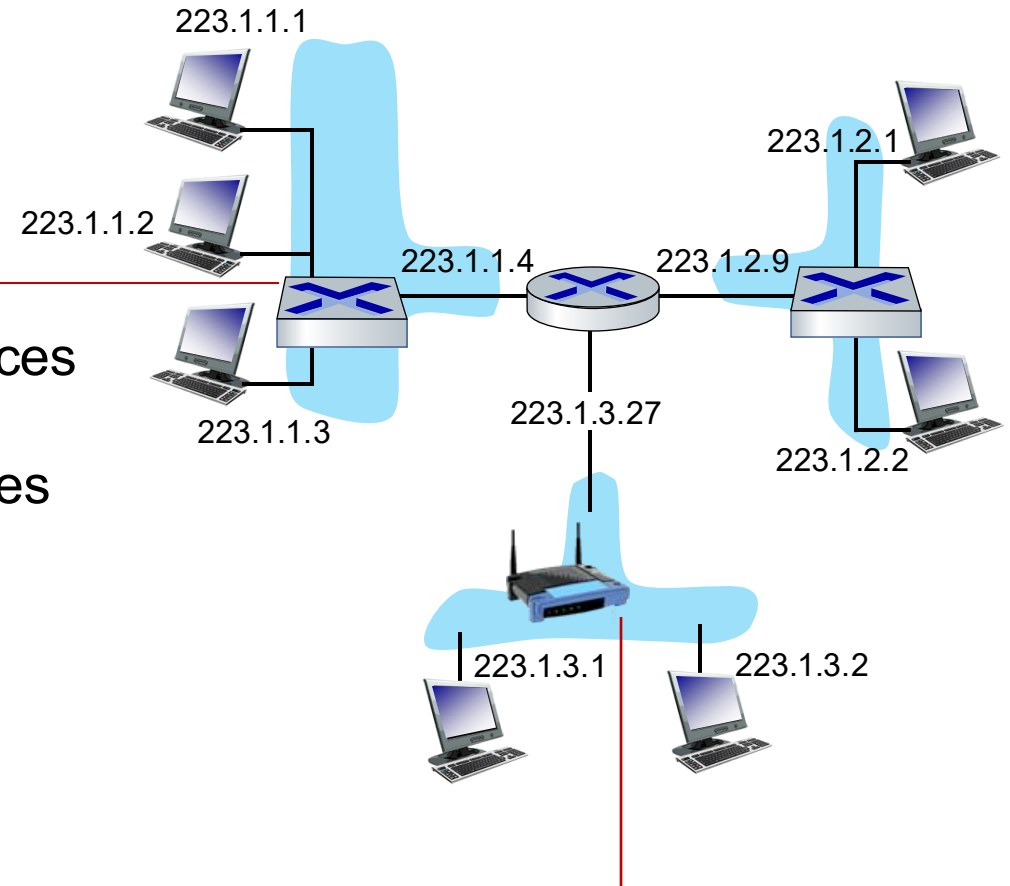
# IP addressing: introduction

**Q:** how are interfaces actually connected?

**A:** we'll learn about that in chapters 6, 7

*For now:* don't need to worry about how one interface is connected to another (with no intervening router)

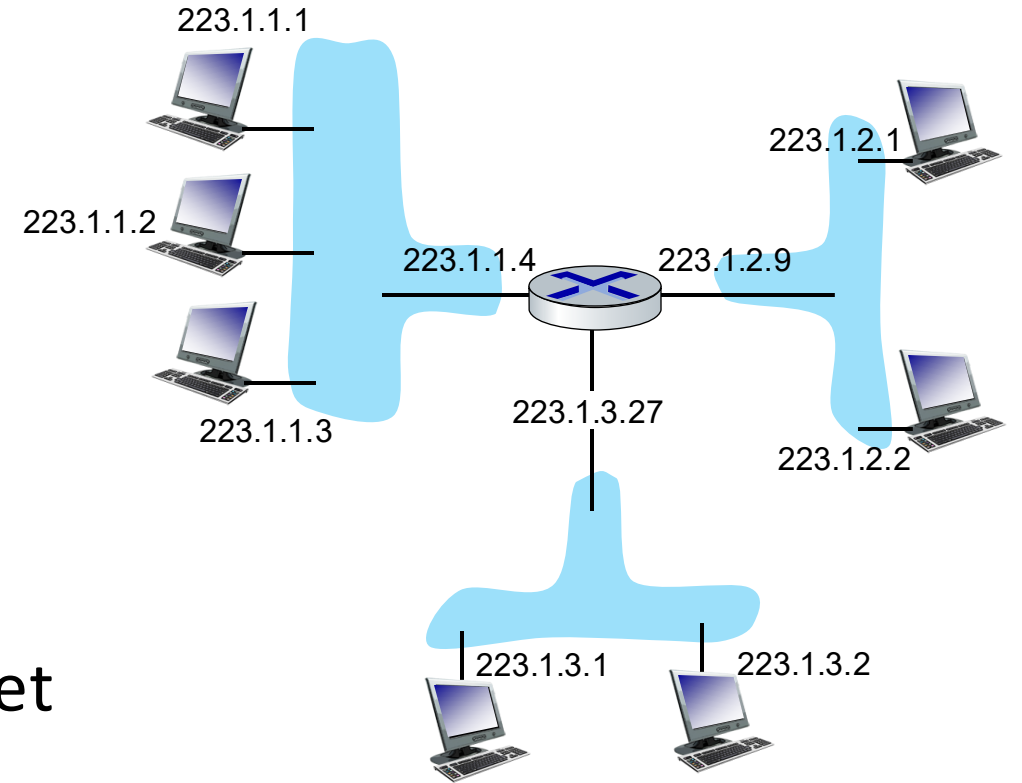
**A:** wired Ethernet interfaces connected by Ethernet switches



**A:** wireless WiFi interfaces connected by WiFi base station

# Subnets

- *What's a subnet ?*
  - device interfaces that can physically reach each other **without passing through an intervening router**
- IP addresses have structure:
  - **subnet part:** devices in same subnet have common high order bits
  - **host part: remaining** low order bits



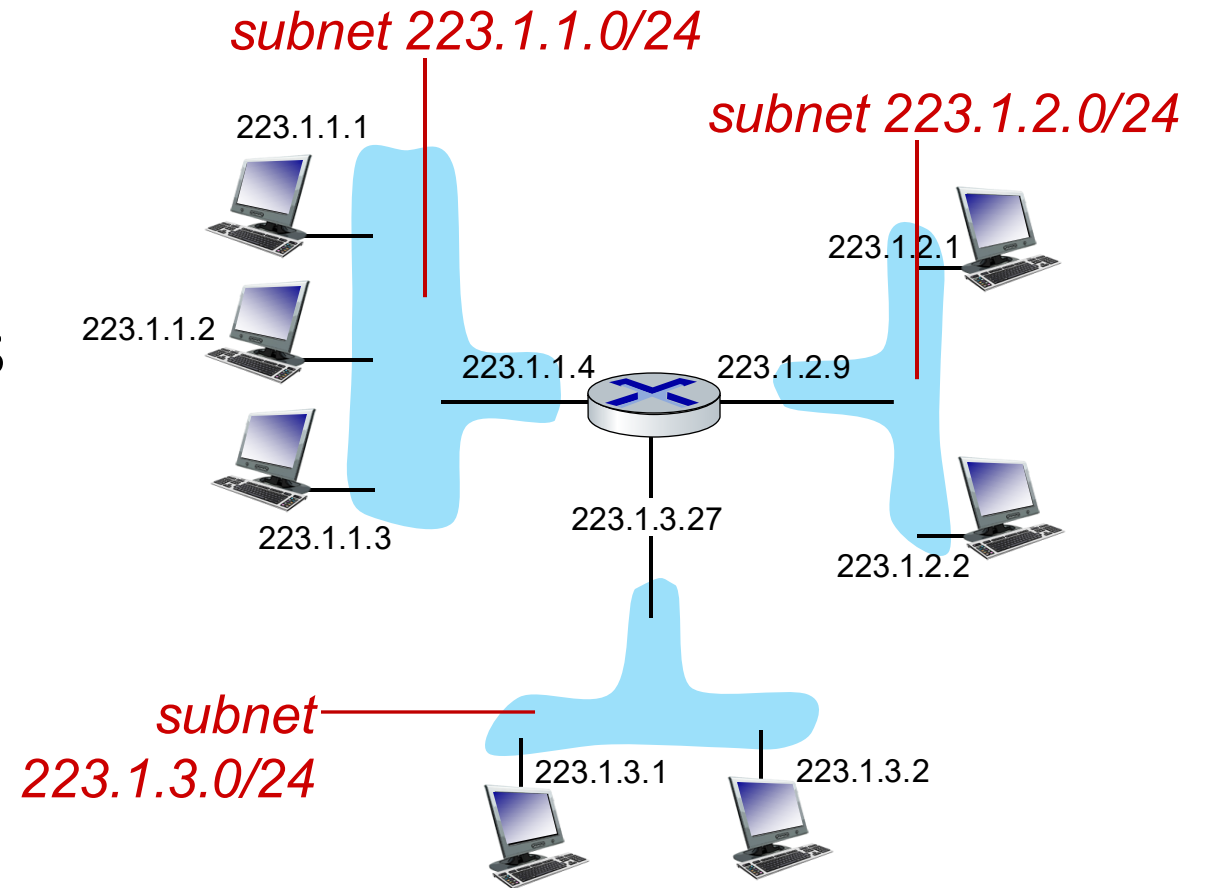
network consisting of 3 subnets



# Subnets

## *Recipe for defining subnets:*

- detach each interface from its host or router, creating “islands” of isolated networks
- each isolated network is called a *subnet*

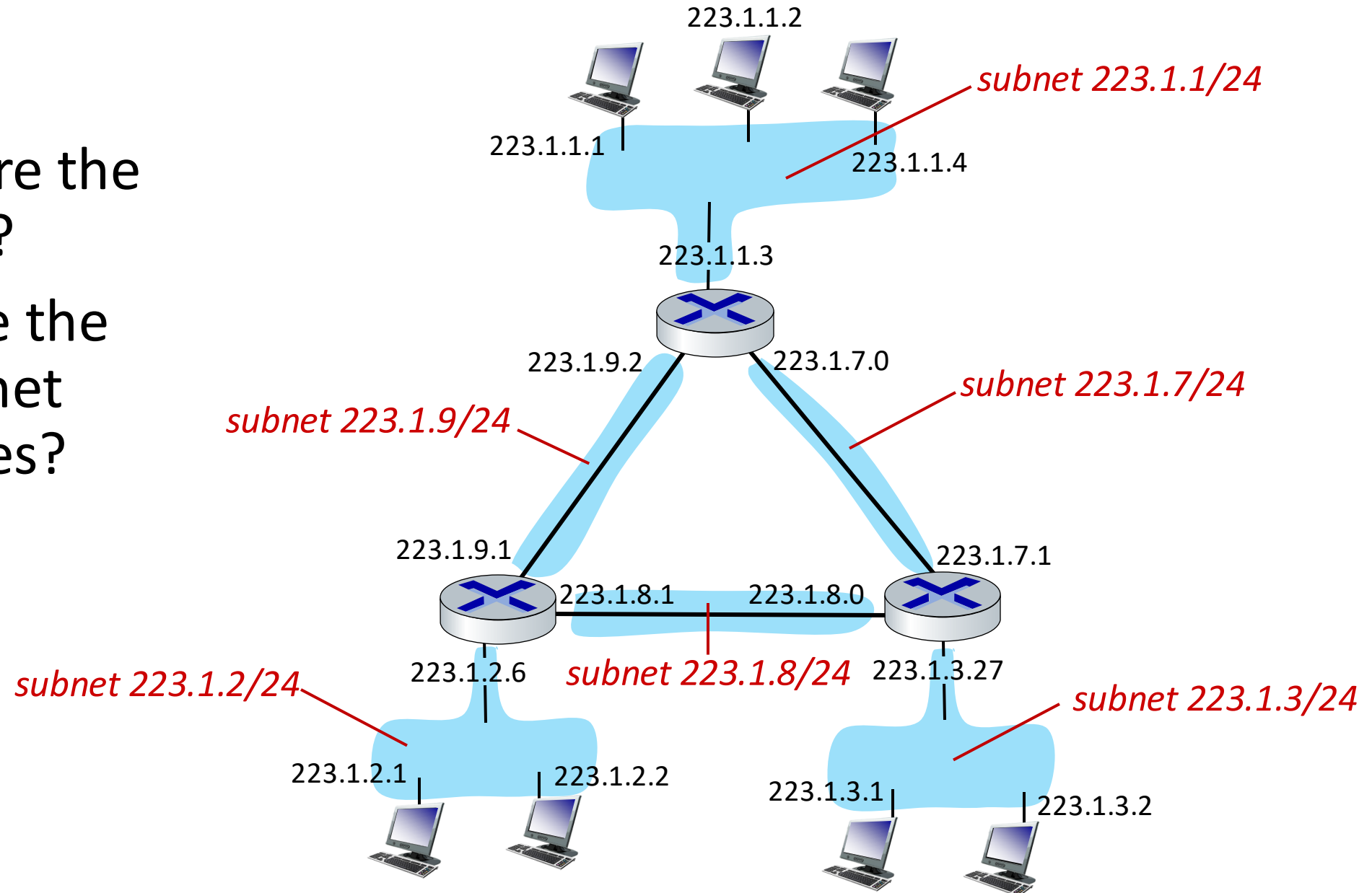


subnet mask: /24

(high-order 24 bits: subnet part of IP address)

# Subnets

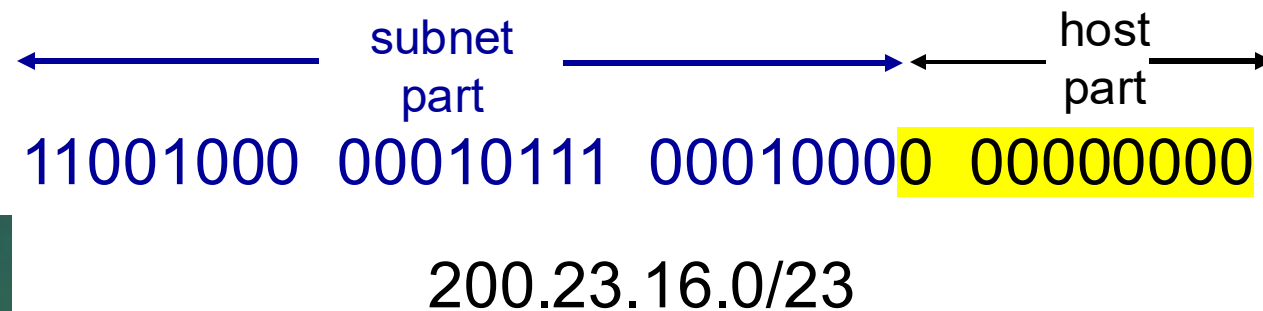
- where are the subnets?
- what are the /24 subnet addresses?



# IP addressing: CIDR

**CIDR: C**lassless **I**nter**D**omain **R**outing (pronounced “cider”)

- subnet portion of address of arbitrary length
- address format: **a.b.c.d/x**, where x is # bits in subnet portion of address



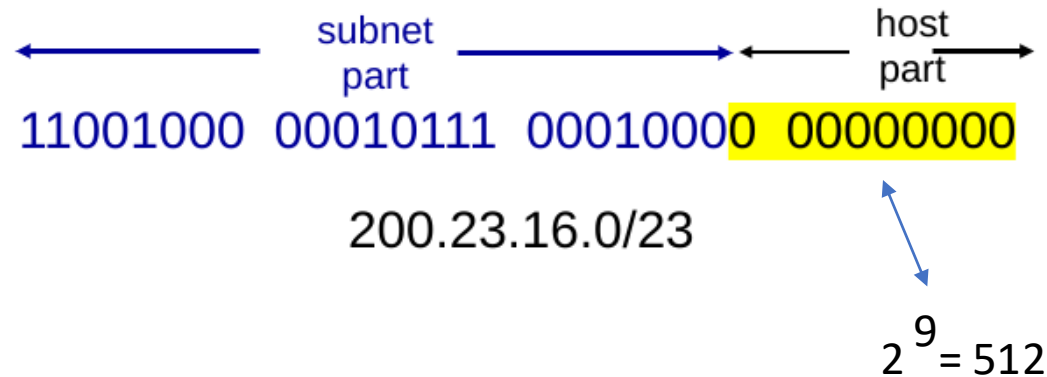
$K = 1,024 \cdot M = 1,048,576$

| IP Addresses | Bits | Prefix | Subnet Mask     |
|--------------|------|--------|-----------------|
| 1            | 0    | /32    | 255.255.255.255 |
| 2            | 1    | /31    | 255.255.255.254 |
| 4            | 2    | /30    | 255.255.255.252 |
| 8            | 3    | /29    | 255.255.255.248 |
| 16           | 4    | /28    | 255.255.255.240 |
| 32           | 5    | /27    | 255.255.255.224 |
| 64           | 6    | /26    | 255.255.255.192 |
| 128          | 7    | /25    | 255.255.255.128 |
| 256          | 8    | /24    | 255.255.255.0   |

# IP addressing: CIDR

$K = 1,024 \cdot M = 1,048,576$

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| 64           | 6    | /26    | 255.255.255.192 |
| 128          | 7    | /25    | 255.255.255.128 |
| 256          | 8    | /24    | 255.255.255.0   |



| Network Bits | Subnet Mask     |
|--------------|-----------------|
| 8            | 255.0.0.0       |
| 9            | 255.128.0.0     |
| 10           | 255.192.0.0     |
| 11           | 255.224.0.0     |
| 12           | 255.240.0.0     |
| 13           | 255.248.0.0     |
| 14           | 255.252.0.0     |
| 15           | 255.254.0.0     |
| 16           | 255.255.0.0     |
| 17           | 255.255.128.0   |
| 18           | 255.255.192.0   |
| 19           | 255.255.224.0   |
| 20           | 255.255.240.0   |
| 21           | 255.255.248.0   |
| 22           | 255.255.252.0   |
| 23           | 255.255.254.0   |
| 24           | 255.255.255.0   |
| 25           | 255.255.255.128 |
| 26           | 255.255.255.192 |
| 27           | 255.255.255.224 |
| 28           | 255.255.255.240 |
| 29           | 255.255.255.248 |
| 30           | 255.255.255.252 |

|     |     |   |    | 4 adresser | 8 adresser | 16 adresser | 32 adresser | 64 adresser | 128 adresser | 256 adresser |
|-----|-----|---|----|------------|------------|-------------|-------------|-------------|--------------|--------------|
|     |     |   |    | /30        | /29        | /28         | /27         | /26         | /25          | /24          |
|     |     |   |    | .252       | .248       | .240        | .224        | .192        | .128         | .0           |
| 192 | 168 | 1 | 0  | -          |            |             |             |             |              |              |
| 192 | 168 | 1 | 1  |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 2  |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 3  | -          |            |             |             |             |              |              |
| 192 | 168 | 1 | 4  |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 5  |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 6  |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 7  |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 8  |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 9  |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 10 |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 11 |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 12 |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 13 |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 14 |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 15 |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 16 |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 17 |            |            |             |             |             |              |              |
| 192 | 168 | 1 | 18 |            |            |             |             |             |              |              |

Første adressen i et subnet er nettverksadressen - tildeles ikke noen  
 Siste adressen i et subnet er broadcast, tildeles heller ikke noen

K = 1,024 • M = 1,048,576

| IP Addresses | Bits | Prefix | Subnet Mask     |
|--------------|------|--------|-----------------|
| 1            | 0    | /32    | 255.255.255.255 |
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| 128          | 7    | /25    | 255.255.255.128 |
| 256          | 8    | /24    | 255.255.255.0   |

# IP addresses: how to get one?

That's actually **two** questions:

1. Q: How does a **host** get IP address within its network (host part of address)?
2. Q: How does a **network** get IP address for itself (network part of address)

How does *host* get IP address?

- hard-coded by sysadmin in config file (e.g., /etc/rc.config in UNIX)
- **DHCP**: Dynamic Host Configuration Protocol: dynamically get address from as server
  - “plug-and-play”

# DHCP: Dynamic Host Configuration Protocol

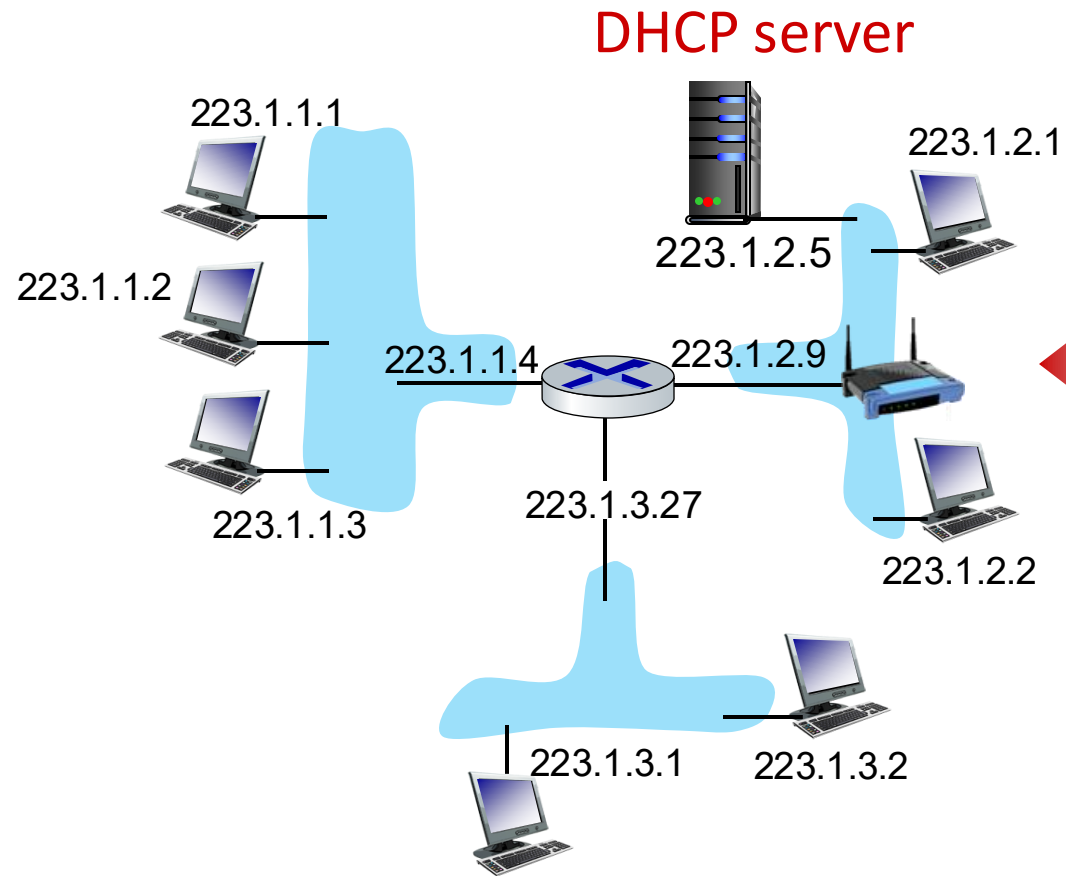
**goal:** host *dynamically* obtains IP address from network server when it “joins” network

- can renew its lease on address in use
- allows reuse of addresses (only hold address while connected/on)
- support for mobile users who join/leave network

## DHCP overview (protocol messages):

- host broadcasts **DHCP discover** msg [optional]
- DHCP server responds with **DHCP offer** msg [optional]
- host requests IP address: **DHCP request** msg
- DHCP server sends address: **DHCP ack** msg

# DHCP client-server scenario



Typically, DHCP server will be co-located in router, serving all subnets to which router is attached

arriving **DHCP client** needs address in this network



# DHCP client-server scenario

DHCP server: 223.1.2.5



**DHCP discover**

Broadcast: is there a  
DHCP server out there?

Arriving client



**DHCP offer**

Broadcast: I'm a DHCP  
server! Here's an IP  
address you can use

**DHCP request**

Broadcast: OK. I would  
like to use this IP address!

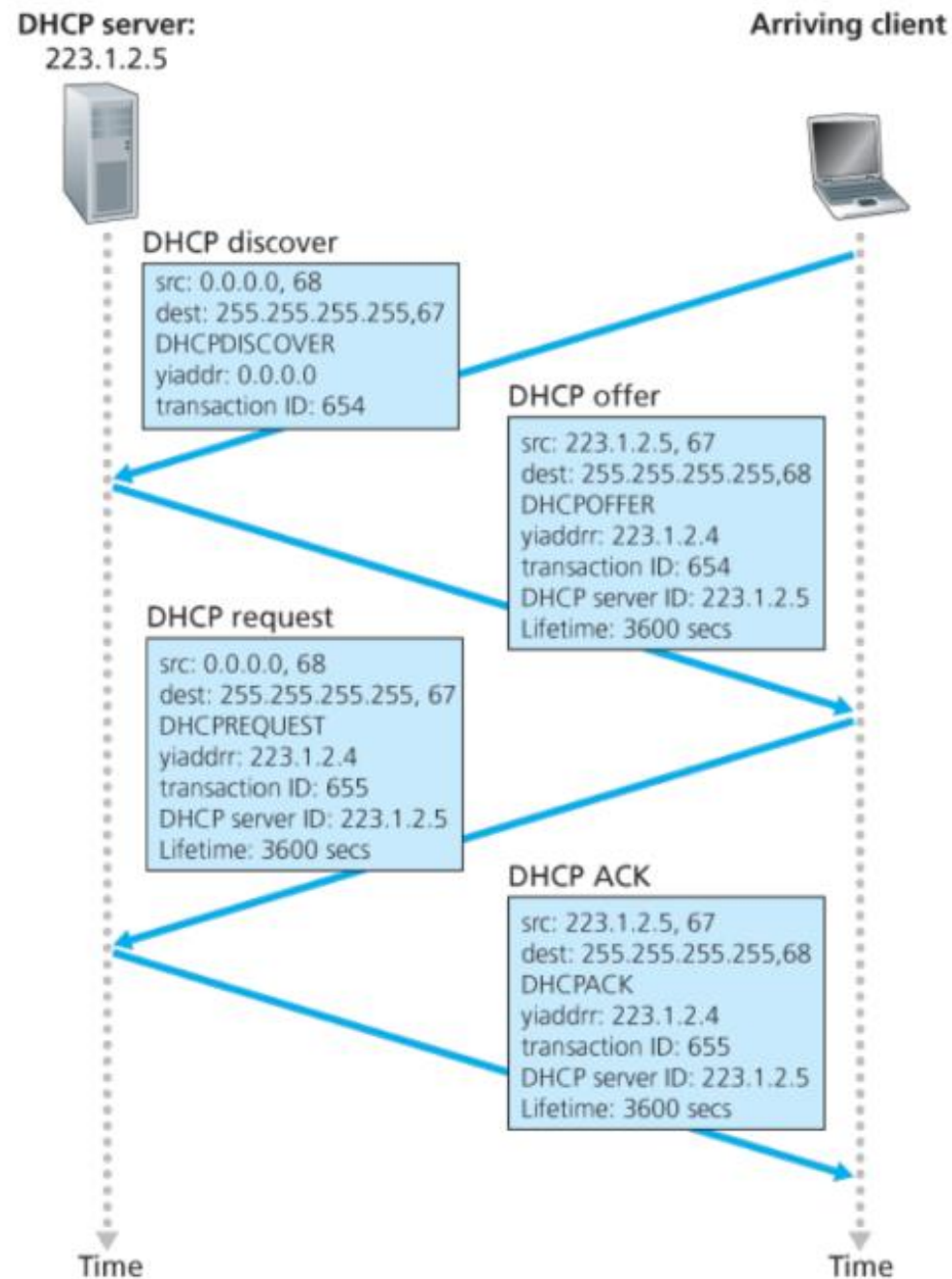
**DHCP ACK**

Broadcast: OK. You've  
got that IP address!

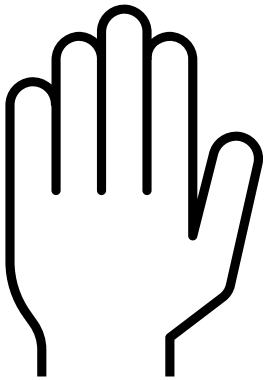
The two steps above can  
be skipped "if a client  
remembers and wishes to  
reuse a previously  
allocated network address"  
[RFC 2131]

The DHCP employs a [connectionless](#) service model, using the [User Datagram Protocol](#) (UDP). It is implemented with two UDP port numbers for its operations which are the same as for the bootstrap protocol ([BOOTP](#)). The server listens on UDP port number 67, and the client listens on UDP port number 68.

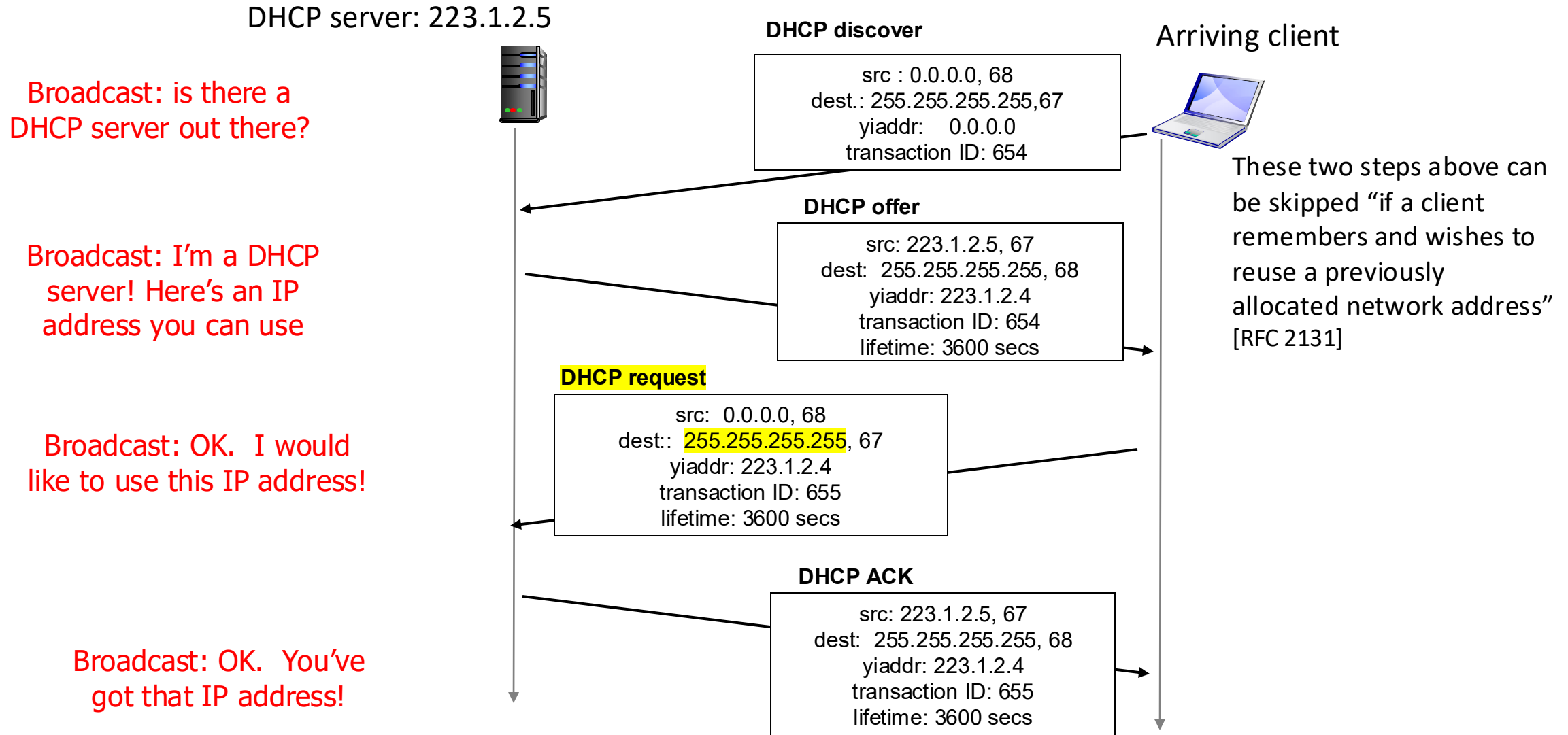
# DHCP client-server ... litt mer detaljer



*Hvorfor sendes DHCPrequest  
melding som broadcast når  
client vet adressen til DHCP  
server I del to av denne  
dialogen ?*



# DHCP client-server scenario



<https://www.rfc-editor.org/rfc/rfc2131>

<https://docs.oracle.com/cd/E19253-01/816-4554/dhcp-overview-3/index.html>



3. The client selects the best offer, based on the number and type of services offered. The client broadcasts a request that specifies the IP address of the server that made the best offer. The broadcast ensures that all the responding DHCP servers know that the client has chosen a server. The servers that are not chosen can cancel the reservations for the IP addresses that they had offered.

3. The client receives one or more DHCP OFFER messages from one or more servers. The client may choose to wait for multiple responses. The client chooses one server from which to request configuration parameters, based on the configuration parameters offered in the DHCP OFFER messages. The client broadcasts a DHCP REQUEST message that MUST include the 'server identifier' option to indicate which server it has selected, and that MAY include other options specifying desired configuration values. The 'requested IP address' option MUST be set to the value of 'yiaddr' in the DHCP OFFER message from the server. This DHCP REQUEST message is broadcast and relayed through DHCP/BOOTP relay agents. To help ensure that any BOOTP relay agents forward the DHCP REQUEST message to the same set of DHCP servers that received the original DHCP DISCOVER message, the DHCP REQUEST message MUST use the same value in the DHCP message header's 'secs' field and be sent to the same IP broadcast address as the original DHCP DISCOVER message. The client times out and retransmits the DHCP DISCOVER message if the client receives no DHCP OFFER messages.
4. The servers receive the DHCP REQUEST broadcast from the client. Those servers not selected by the DHCP REQUEST message use the message as notification that the client has declined that server's offer. The server selected in the DHCP REQUEST message commits the binding for the client to persistent storage and responds with a DHCP ACK message containing the configuration parameters for the requesting client. The combination of 'client identifier' or 'chaddr' and assigned network address constitute a unique identifier for the client's lease and are used by both the client and server to identify a lease referred to in any DHCP messages. Any configuration parameters in the DHCP ACK message SHOULD NOT conflict with those in the earlier DHCP OFFER message to which the client is responding. The server SHOULD NOT check the offered network address at this point. The 'yiaddr' field in the DHCP ACK messages is filled in with the selected network address.

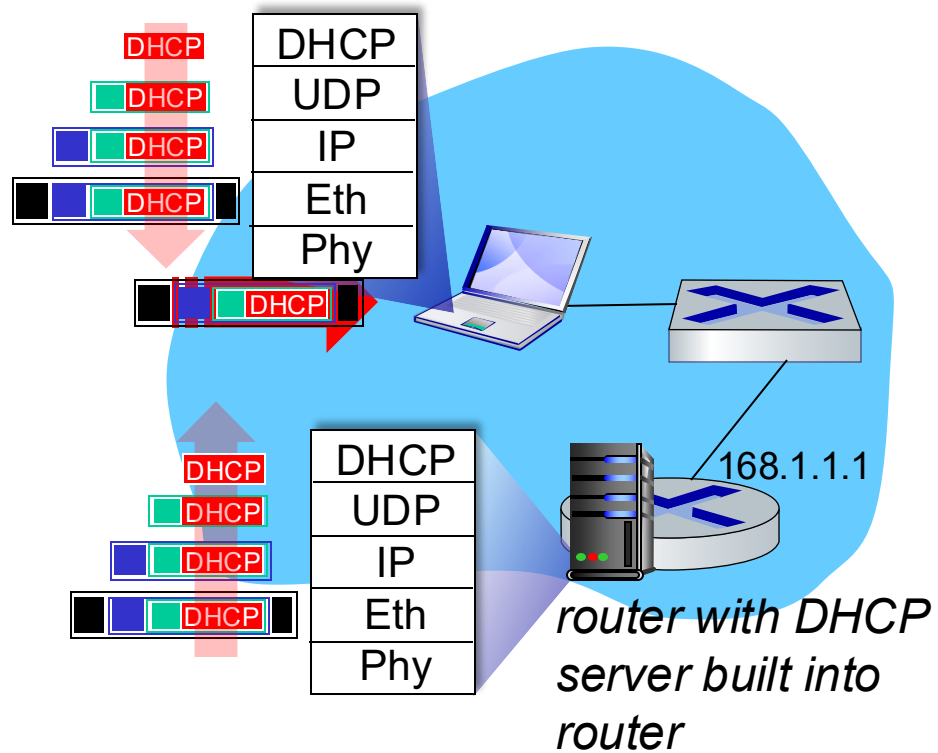
# DHCP: more than IP addresses

DHCP can return more than just allocated IP address on subnet:

- address of first-hop router for client
- name and IP address of DNS sever
- network mask (indicating network versus host portion of address)



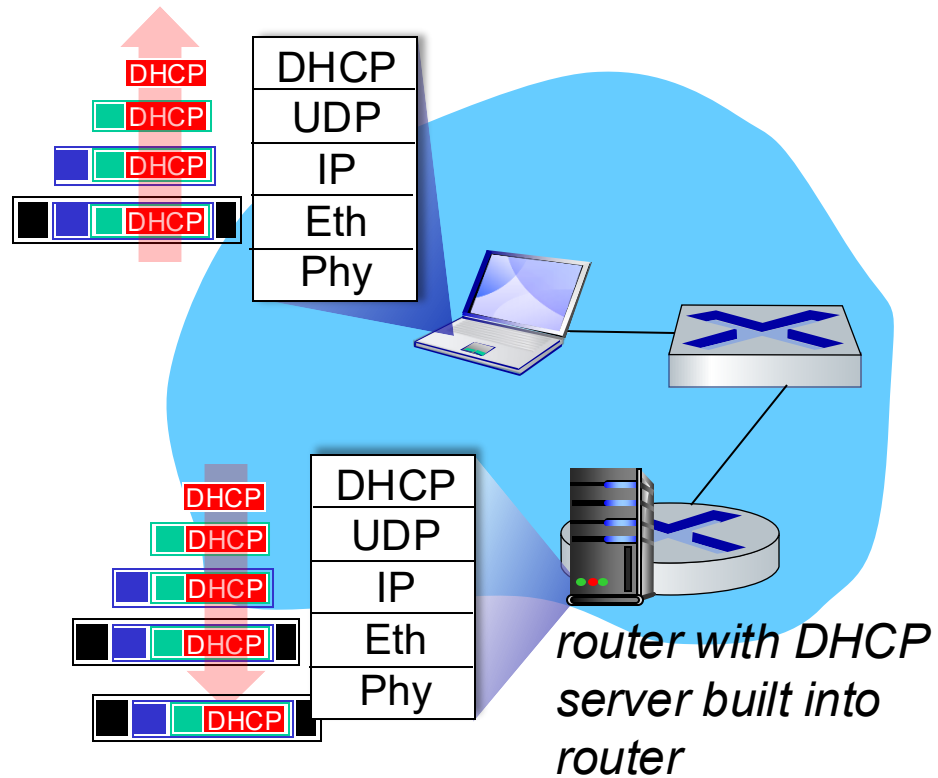
# DHCP: example



- Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server.
- DHCP REQUEST message encapsulated in UDP, encapsulated in IP, encapsulated in Ethernet
- Ethernet frame broadcast (dest: FFFFFFFF) on LAN, received at router running DHCP server
- Ethernet demux'ed to IP demux'ed, UDP demux'ed to DHCP



# DHCP: example



- DCP server formulates DHCP ACK containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- encapsulated DHCP server reply forwarded to client, demuxing up to DHCP at client
- client now knows its IP address, name and IP address of DNS server, IP address of its first-hop router

# IP addresses: how to get one?

**Q:** how does *network* get subnet part of IP address?

**A:** gets allocated portion of its provider ISP's address space

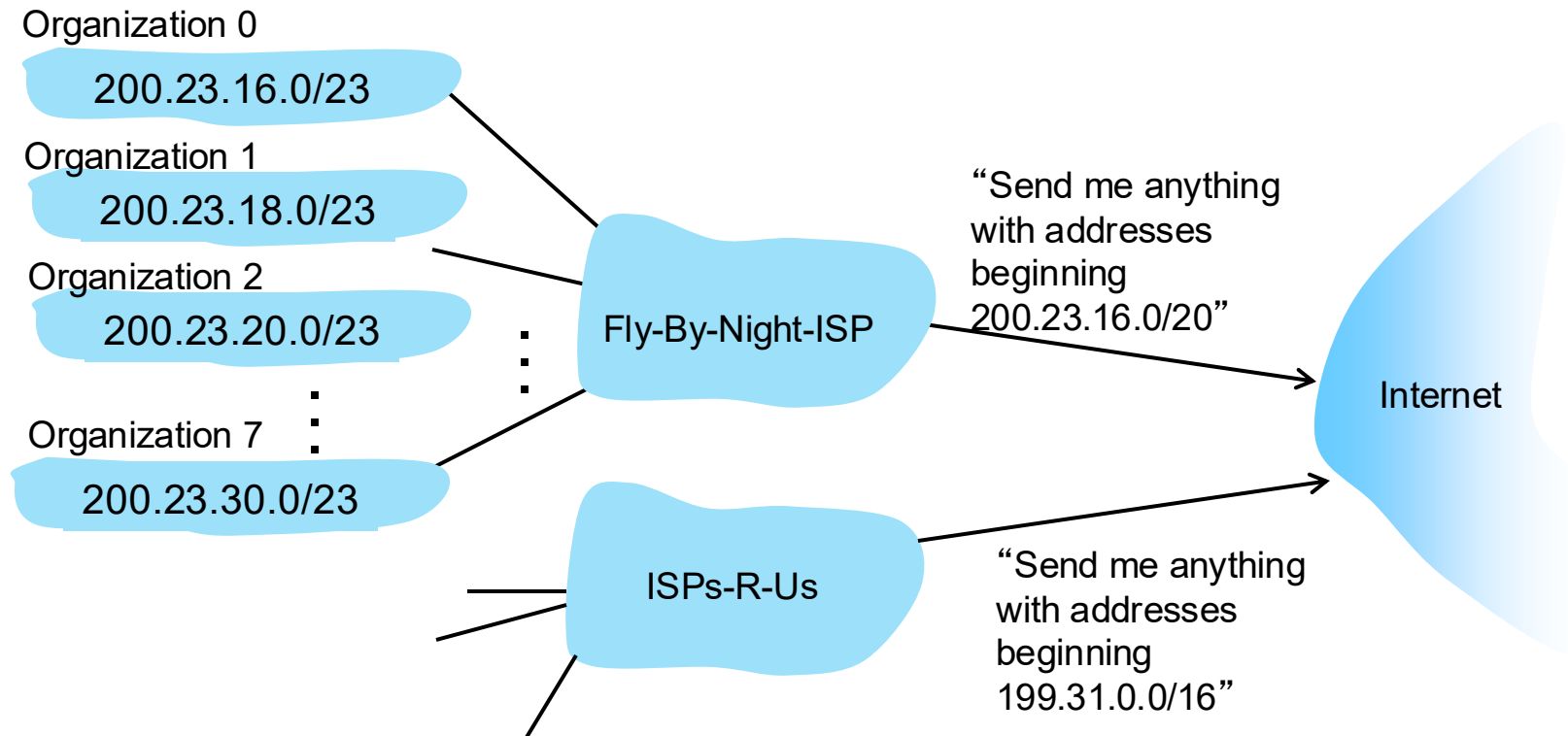
ISP's block      11001000 00010111 00010000 00000000    200.23.16.0/20

ISP can then allocate out its address space in 8 blocks:

|                |                                   |          |                |
|----------------|-----------------------------------|----------|----------------|
| Organization 0 | <u>11001000 00010111 00010000</u> | 00000000 | 200.23.16.0/23 |
| Organization 1 | <u>11001000 00010111 00010010</u> | 00000000 | 200.23.18.0/23 |
| Organization 2 | <u>11001000 00010111 00010100</u> | 00000000 | 200.23.20.0/23 |
| ...            | .....                             | ....     | ....           |
| Organization 7 | <u>11001000 00010111 00011110</u> | 00000000 | 200.23.30.0/23 |

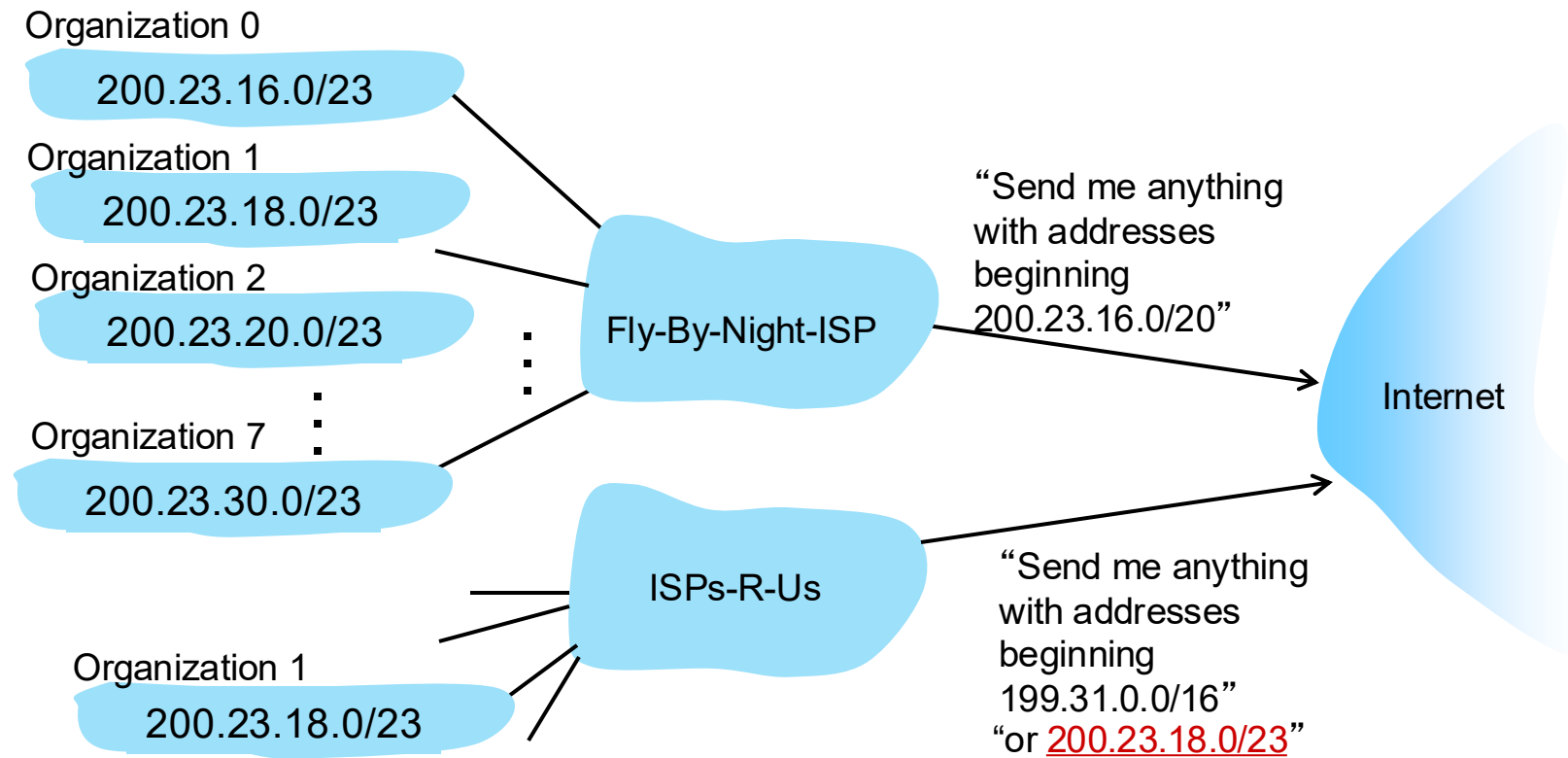
# Hierarchical addressing: route aggregation

hierarchical addressing allows efficient advertisement of routing information:



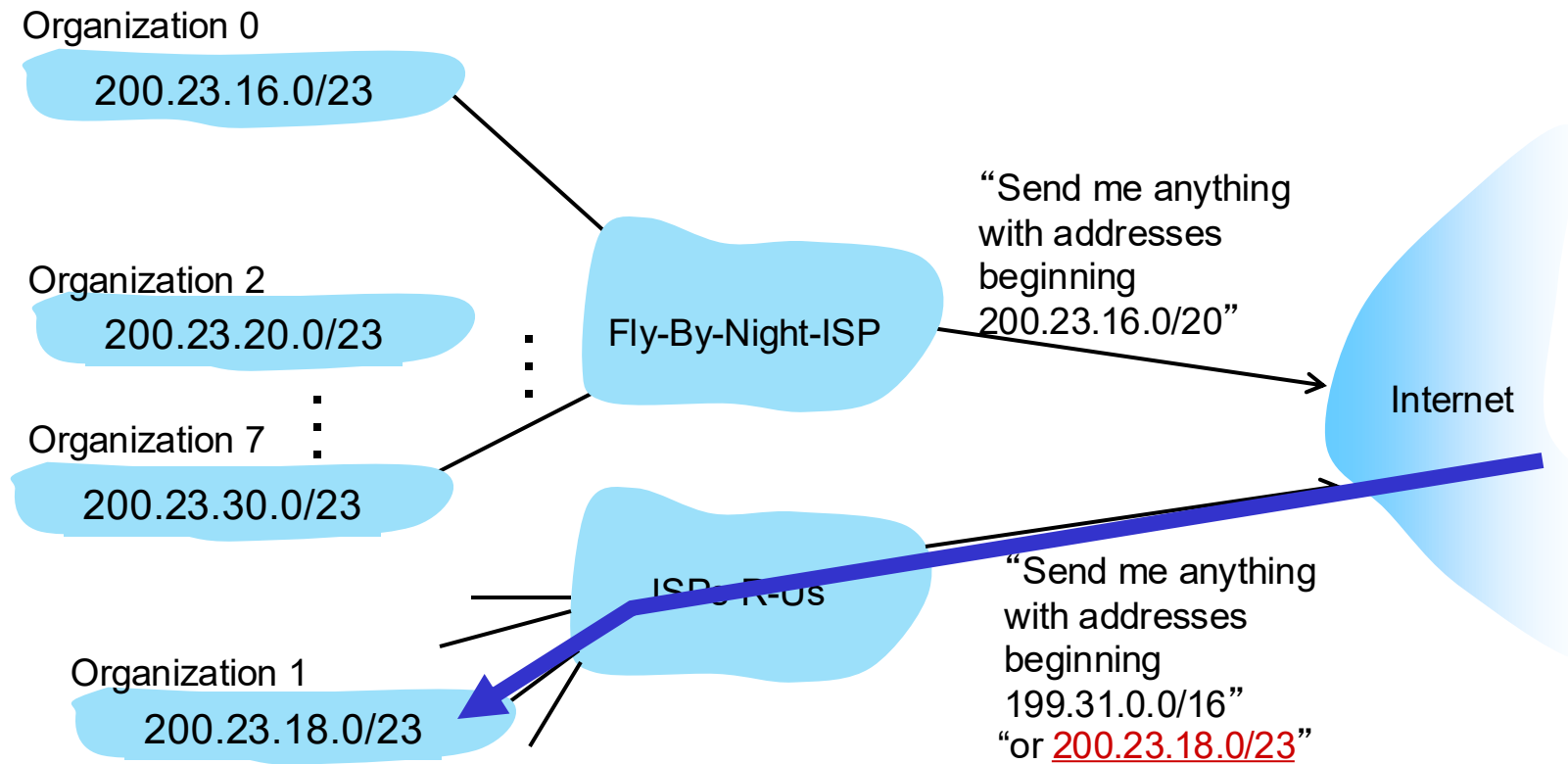
# Hierarchical addressing: more specific routes

- Organization 1 moves from Fly-By-Night-ISP to ISPs-R-Us
- ISPs-R-Us now advertises a more specific route to Organization 1



# Hierarchical addressing: more specific routes

- Organization 1 moves from Fly-By-Night-ISP to ISPs-R-Us
- ISPs-R-Us now advertises a more specific route to Organization 1



# IP addressing: last words ...

**Q:** how does an ISP get block of addresses?

**A:** ICANN: Internet Corporation for Assigned Names and Numbers  
<http://www.icann.org/>

- allocates IP addresses, through 5 regional registries (RRs) (who may then allocate to local registries)
- manages DNS root zone, including delegation of individual TLD (.com, .edu , ...) management

**Q:** are there enough 32-bit IP addresses?

- ICANN allocated last chunk of IPv4 addresses to RRs in 2011
- NAT (next) helps IPv4 address space exhaustion
- IPv6 has 128-bit address space

"Who the hell knew how much address space we needed?" Vint Cerf (reflecting on decision to make IPv4 address 32 bits long)